

MUSCLING IN ON OZEMPIC AND MOUNJARO: THE IMPACT OF THE NEW GENERATION OF GLUCAGON-LIKE PEPTIDE RECEPTOR AGONISTS ON MUSCLE BIOLOGY

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INTRODUCTION

Glucagon-like peptide receptor agonists (GLP-1RAs) have become a popular tool for weight loss and glycaemic reduction, however, there is limited research on the direct impact of these pharmacological agents on human muscle cells. In human trials there are mixed results, with a range of 8 to 46% of the weight loss coming from lean mass (1-7). It is currently unclear if this is because of GLP-1RAs, nutrition, exercise or reduced loading on the muscles. *In vitro* and *in vivo*, previously approved GLP-1RAs have actually shown beneficial effects to the muscle and cells by reducing atrophy markers, inflammation and attenuating muscle loss (8-12). Because of this, further testing in a controlled environment, without mechanistic inputs should be considered.

Using a human cell culture model creates a controlled environment that can show the direct effects of GLP-1RAs on the muscle cells, without the confounding factors discussed above (13). Morphological changes to the myotubes, including fibre size and maturation rate, can show the direct effects of GLP-1RAs on the treated cells (14, 15). However, there is limited information about how recently approved GLP-1RAs, Semaglutide and Tirzepatide, affect these factors in human muscle cells. Overall, further information about the effects of GLP-1RAs on the muscle is still required as there are concerns in the community and scientists that these medications exacerbate the loss of muscle mass.

METHODS

Pilot study: Previous studies have used a range of doses across the different GLP-1RAs. As such, it is necessary to first establish the optimal dose for muscle growth without causing toxicity (16). A 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide assay (MTT) was conducted to establish the optimal dose of Semaglutide and Tirzepatide. Human primary muscle cells (n=2) were cultured under normal conditions until differentiation. Concentrations between 1 to 600nM of each treatment were applied to muscle cells for 48 hours and subsequently read using a plate reader.

Main study: Once the optimal dose of 50nM was identified, human primary muscle cells (n=4) were cultured under normal conditions until differentiation. During differentiation, cells were either assigned to Semaglutide, Tirzepatide or vehicle treatment in a 1:1:1 ratio. Morphological measures were collected at day four and day seven of differentiation by staining the cells and analysed using confocal microscopy to obtain images.

RESULTS AND DISCUSSION

Pilot study: Figure 1 shows the viability of the cells, which established that the optimal dose would be 20 to 50nM of either treatment.

Main study: This study is still being conducted.

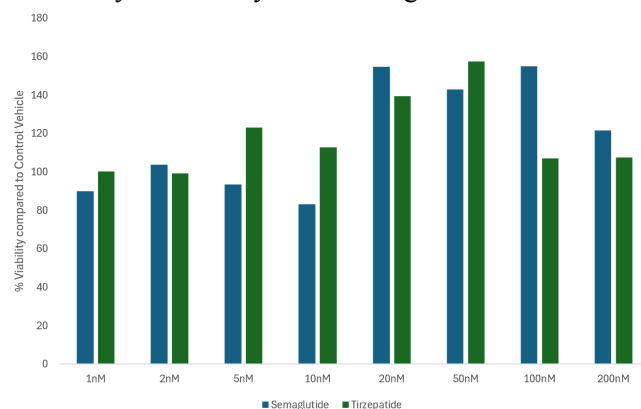


Figure 1 Pilot study showing the viability of human primary muscle cells when treated with either Semaglutide or Tirzepatide compared to a vehicle control.

CONCLUSIONS

The main study is still being conducted, with results and discussion to be presented at the symposium.

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